# Team NOOBLERS

Electric Fixed-Wing UAV

Project Technical Report

Laws of Motion | Kshitij 2025, IIT Kharagpur

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| **Team Name** | NOOBLERS |
| **Team Members** | Soumyadip Das (Team Leader), Ankan Prasad Roy, Aparna Dutta, Souptik De |
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| **Event** | Laws of Motion, Kshitij 2025 \| IIT Kharagpur |
| **RC Frequency** | 2.4 GHz (ELRS) |

## 1. Abstract

This report presents the complete technical design, fabrication methodology, propulsion analysis, avionics integration, and performance characterisation of Team NOOBLERS' entry for the Laws of Motion aerial robotics competition at Kshitij 2025, IIT Kharagpur. The aircraft is a lightweight, electric, fixed-wing platform engineered for stable low-speed flight, reliable payload delivery, and ease of on-field maintenance. Every design decision — from airfoil selection to material choice — was guided by the dual imperatives of maximising aerodynamic efficiency and minimising all-up weight within the constraints of readily available, low-cost components.


## 2. Team Information

Team NOOBLERS is a four-member team comprising Soumyadip Das (Team Leader), Ankan Prasad Roy, Aparna Dutta, and Souptik De, competing under the Laws of Motion event at Kshitij 2025, IIT Kharagpur.

## 3. Introduction and Design Philosophy

The Laws of Motion competition challenges teams to design, fabricate, and fly a fixed-wing aircraft capable of completing a series of mission tasks with precision and reliability. The design philosophy of Team NOOBLERS centres on three guiding principles:

- **Aerodynamic Stability:** Prioritising gentle stall behaviour and wide speed margins to handle turbulence and pilot error gracefully.
- **Structural Simplicity:** Employing foam-composite construction to enable rapid prototyping, field repair, and weight reduction without sacrificing rigidity.
- **System Reliability:** Selecting proven commercial off-the-shelf propulsion and avionics components to minimise failure modes during competition.

The resulting aircraft achieves an all-up weight of approximately 790 g, a thrust-to-weight ratio of 1.1:1, and a predicted stall speed of 7–8 m/s, placing it firmly within the safe-flight envelope required by the competition.

## 4. Aerodynamic Design

### 4.1 Airfoil Selection

The Selig S1223 high-lift airfoil was selected following a comparative analysis of low-Reynolds-number profiles. The S1223 is a cambered section purpose-designed for model aircraft and small UAVs operating in the Re = 50,000–300,000 range. Its principal characteristics are:

- Maximum $C_L$ of approximately 2.2, substantially higher than symmetric or lightly cambered profiles.
- Gradual stall onset: lift degrades progressively beyond $C_{L,max}$, providing additional pilot recovery time.
- Moderate pitching moment ($C_m$ approximately -0.13), compatible with a conventional tail volume without excessive trim drag.
- Well-documented performance data enabling reliable analytical predictions.

### 4.2 Wing Geometry

| Parameter | Value |
|---|---|

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|---|---|
| Wingspan | 1160 mm |
| Chord Length (constant) | 270 mm |
| Wing Area | 0.313 $m^2$ |
| Aspect Ratio | 4.30 |
| Dihedral | 3° (estimated for lateral stability) |
| Airfoil | Selig S1223 |
| Wing Loading at 790 g AUW | ~24.8 $N/m^2$ |

The unswept, constant-chord (rectangular planform) wing was chosen to simplify hot-wire cutting and maximise wing area for a given span. An aspect ratio of 4.3 balances induced drag reduction against structural bending stiffness.

### 4.3 Estimated Aerodynamic Performance

| Metric | Estimated Value | Remarks |
|---|---|---|
| Stall Speed | 7–8 m/s | $C_{L,max} \approx 2.0$ assumed |
| Cruise Speed | 10–12 m/s | Optimal L/D region |
| Maximum Speed | ~18 m/s | Propulsion limited |
| Lift-to-Drag Ratio | ~8–10 | At cruise angle of attack |
| Cruise Reynolds Number | ~190,000 | Based on 270 mm chord |

## 5. Structural Design and Fabrication

### 5.1 Wing Structure

Wing panels are fabricated from expanded polystyrene (EPS) styrofoam using a template-guided hot-wire cutter, achieving sub-millimetre fidelity to the S1223 profile at a panel density of approximately 15–20 $kg/m^3$.

- **Surface finishing:** A thin layer of diluted PVA or fibre-reinforced tape is applied to improve rigidity, resist leading-edge abrasion, and provide a smooth aerodynamic finish.
- **Spar:** A carbon-fibre rod runs spanwise at the 25% chord line to resist bending loads and transmit control surface loads.
- **Ailerons:** Cut from the trailing edge and hinged with Mylar tape for minimal play.

### 5.2 Fuselage Structure

The fuselage is a hybrid corosheet (corrugated polypropylene) and styrofoam box structure, approximately 850 mm in overall length. Corosheet provides compressive strength along the corrugation direction and resists impact deformation — important for recovery from uncontrolled landings.

- **Nose section:** Carved styrofoam for crash energy absorption.
- **Centre section:** Corosheet box housing the battery, ESC, and receiver on Velcro mounts for rapid field replacement.
- **Tail boom:** Integrated corosheet extension carrying the horizontal and vertical stabilisers.

- **Payload bay:** Dedicated compartment with a servo-actuated door for competition payload release tasks.

### 5.3 Tail Assembly

A conventional cruciform tail provides pitch and yaw stability. The tail moment arm of approximately 520 mm, combined with appropriately sized stabiliser areas, yields an estimated static margin of 5–10% MAC — within the recommended range for stable, easily pilotable aircraft.

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## 6. Propulsion System

The propulsion system was selected to deliver adequate thrust for short-field takeoff and climb while remaining within the current and thermal limits of a 3S lithium-polymer battery.

| Component | Specification | Remarks |
|---|---|---|
| Brushless Motor | 2212 / 1400 KV | High efficiency at 3S voltage |
| Battery | 3S 2200 mAh Li-Po | 11.1 V nominal |
| Electronic Speed Controller | 30 A | 12 A headroom over cruise draw |
| Propeller | 10×4.5 twin-blade | ~870 g static thrust |
| Thrust-to-Weight Ratio | ~1.10 : 1 | Safe short-field takeoff |
| No-load Motor Speed (3S) | ~15,540 RPM | KV × nominal voltage |
| Estimated Flight Current | 12–18 A (cruise) | ~80% of ESC rating |
| Estimated Flight Duration | 12–16 min | At 50% average throttle |

The 1045 propeller is an optimal match for the 2212/1400 KV motor on 3S power. Its larger diameter (10 inch) at moderate pitch (4.5 inch) maximises static thrust efficiency during the low-speed takeoff phase while keeping tip speeds below compressibility thresholds. The 30 A ESC provides a 12 A safety margin over expected cruise current, preventing thermal shutdown during sustained full-throttle climbs.

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## 7. Avionics and Control System

### 7.1 Transmitter and Receiver

The aircraft uses a Radiomaster Pocket transmitter paired with a PWM-based ExpressLRS (ELRS) receiver operating at 2.4 GHz. ELRS is an open-source, low-latency protocol offering sub-5 ms control latency and a link budget exceeding 120 dB, providing robust and responsive operation in all expected competition environments.

### 7.2 Servo Allocation

| Channel | Control Surface / Function | Servo Type |
|---|---|---|
| 1 | Left Aileron | 9 g micro servo |
| 2 | Right Aileron | 9 g micro servo |
| 3 | Elevator | 9 g micro servo |
| 4 | Rudder | 9 g micro servo |

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|---|---|---|
| 5 | Payload Bay Door | 9 g micro servo |

All five servos deliver approximately 1.5 kg·cm torque at 4.8 V. Control throws are configured conservatively (±15° for ailerons and elevator, ±20° for rudder) with dual-rate switches available on the transmitter.

### 7.3 Power Distribution

The 3S Li-Po feeds the ESC directly. The ESC's integrated Battery Elimination Circuit (5 V / 3 A) supplies the receiver and all five servos, eliminating the need for a separate BEC or additional power rail.

## 8. Weight and Balance

| Component | Estimated Mass (g) |
|---|---|
| Wing (foam + surface finish + spar) | 160 |
| Fuselage (corosheet + foam) | 120 |
| Tail assembly | 45 |
| Motor (2212) | 55 |
| ESC (30 A) | 28 |
| Battery (3S 2200 mAh Li-Po) | 185 |
| Receiver (ELRS) | 8 |
| 5 × Servo (9 g each) | 45 |
| Propeller and motor mount | 35 |
| Wiring, connectors, and hardware | 30 |
| Payload | 79 |
| **Total All-Up Weight** | **~790 g** |

The centre of gravity is positioned at approximately 90 mm (33% chord) aft of the leading edge, slightly forward of the neutral point to achieve positive static margin. The battery is used as the primary CG-adjustment mass and is mounted on a sliding rail to allow fine-tuning without structural modification.

## 9. Performance Summary

| Parameter | Value | Unit |
|---|---|---|
| All-Up Weight (AUW) | 790 | g |
| Maximum Static Thrust | 870 | g |
| Thrust-to-Weight Ratio | 1.10 | — |
| Wing Area | 0.313 | $m^2$ |
| Wing Loading | 24.8 | $N/m^2$ |

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|---|---|---|
| Stall Speed ($V_S$) | 7–8 | m/s |
| Cruise Speed | 10–12 | m/s |
| Maximum Speed | ~18 | m/s |
| Estimated Endurance | 12–16 | min |
| RC Link Frequency | 2.4 | GHz |
| RC Protocol | ELRS (PWM) | — |

## 10. Safety Measures and Testing Protocol

Ground and flight testing is conducted at each stage of fabrication to verify structural integrity, control authority, and propulsion reliability prior to competition.

- **Structural load test:** Wing panels are loaded to 3g (2.4 kg distributed) to confirm deflection is within elastic limits and that no delamination occurs.
- **Control linkage check:** All servo horns, pushrods, and control surfaces are verified for zero play and correct direction of travel before every flight.
- **Motor and ESC burn-in:** Full-throttle run for two minutes on the bench with temperature monitoring of motor windings and ESC.
- **Range check:** ELRS link verified at over 200 m on the ground with the motor running to confirm RF noise immunity.
- **CG verification:** Aircraft balanced at the 33% chord line before every flight; battery position adjusted as required.
- **Maiden flight protocol:** Low-altitude straight-and-level pass to verify trim, followed by gradual envelope expansion over three to four flights.
- **Emergency cut:** A dedicated transmitter switch cuts the motor to zero throttle instantly in the event of an emergency.

## 11. Conclusion

The aircraft designed by Team NOOBLERS represents a carefully optimised, competition-ready electric fixed-wing platform. The selection of the Selig S1223 high-lift airfoil, combined with a lightweight foam-composite airframe, a well-matched 2212/1400 KV propulsion system, and a modern ELRS digital radio link, results in an aircraft with favourable stall margins, a generous thrust-to-weight ratio, and reliable control responsiveness.

The team is confident that the aircraft meets the performance requirements of the Laws of Motion competition and is prepared for a strong performance at Kshitij 2025, IIT Kharagpur. Ongoing refinement based on flight test data will further optimise the centre of gravity position, control throws, and propeller pitch to maximise mission performance on competition day.

Submitted by Team NOOBLERS | soumyadipdas1710@gmail.com | Kshitij 2025, IIT Kharagpur